

01_eda

April 17, 2026

CERT Dataset — Exploratory Data Analysis

Import Libraries

```
[2]: import sys
import os

# Add project root directory to path
sys.path.append(os.path.abspath(".."))
```

```
[3]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from src.config import config

%matplotlib inline
```

Load Processed Data

```
[4]: logs = {
    name: pd.read_parquet(config.PROCESSED_DIR / f"{name}_processed.parquet")
    for name in config.LOG_FILES
}
```

Basic Overview

```
[5]: for name, df in logs.items():
    print(f"\n{name.upper()}")
    print("Shape:", df.shape)
    print("Columns:", df.columns.tolist())
    print("Nulls:\n", df.isnull().sum())
```

LOGON

Shape: (854859, 11)

Columns: ['id', 'user', 'pc', 'activity', 'timestamp', 'hour', 'day_of_week', 'is_weekend', 'is_after_hours', 'date_only', 'log_source']

Nulls:

id	0
user	0

```

pc                0
activity          0
timestamp         0
hour              0
day_of_week       0
is_weekend        0
is_after_hours    0
date_only         0
log_source        0
dtype: int64

```

DEVICE

Shape: (405380, 11)

Columns: ['id', 'user', 'pc', 'activity', 'timestamp', 'hour', 'day_of_week', 'is_weekend', 'is_after_hours', 'date_only', 'log_source']

Nulls:

```

id                0
user              0
pc                0
activity          0
timestamp         0
hour              0
day_of_week       0
is_weekend        0
is_after_hours    0
date_only         0
log_source        0
dtype: int64

```

EMAIL

Shape: (2629979, 17)

Columns: ['id', 'user', 'pc', 'to', 'cc', 'bcc', 'from', 'size', 'attachments', 'content', 'timestamp', 'hour', 'day_of_week', 'is_weekend', 'is_after_hours', 'date_only', 'log_source']

Nulls:

```

id                0
user              0
pc                0
to                0
cc                1617054
bcc                2212977
from              0
size              0
attachments       0
content           0
timestamp         0
hour              0
day_of_week       0

```

```
is_weekend          0
is_after_hours       0
date_only            0
log_source           0
dtype: int64
```

FILE

```
Shape: (445581, 12)
```

```
Columns: ['id', 'user', 'pc', 'filename', 'content', 'timestamp', 'hour',
'day_of_week', 'is_weekend', 'is_after_hours', 'date_only', 'log_source']
```

Nulls:

```
id          0
user        0
pc          0
filename    0
content     0
timestamp   0
hour        0
day_of_week 0
is_weekend  0
is_after_hours 0
date_only   0
log_source  0
dtype: int64
```

HTTP

```
Shape: (28434423, 12)
```

```
Columns: ['id', 'user', 'pc', 'url', 'content', 'timestamp', 'hour',
'day_of_week', 'is_weekend', 'is_after_hours', 'date_only', 'log_source']
```

Nulls:

```
id          0
user        0
pc          0
url         0
content     0
timestamp   0
hour        0
day_of_week 0
is_weekend  0
is_after_hours 0
date_only   0
log_source  0
dtype: int64
```

Null Analysis

```
[6]: for name, df in logs.items():
      print(f"\n{name}")
```

```
print((df.isnull().sum() / len(df) * 100).round(2))
```

```
logon
id          0.0
user        0.0
pc          0.0
activity    0.0
timestamp   0.0
hour        0.0
day_of_week 0.0
is_weekend  0.0
is_after_hours 0.0
date_only   0.0
log_source  0.0
dtype: float64
```

```
device
id          0.0
user        0.0
pc          0.0
activity    0.0
timestamp   0.0
hour        0.0
day_of_week 0.0
is_weekend  0.0
is_after_hours 0.0
date_only   0.0
log_source  0.0
dtype: float64
```

```
email
id          0.00
user        0.00
pc          0.00
to          0.00
cc          61.49
bcc         84.14
from        0.00
size        0.00
attachments 0.00
content     0.00
timestamp   0.00
hour        0.00
day_of_week 0.00
is_weekend  0.00
is_after_hours 0.00
date_only   0.00
```

```
log_source      0.00
dtype: float64
```

```
file
id      0.0
user    0.0
pc      0.0
filename 0.0
content 0.0
timestamp 0.0
hour     0.0
day_of_week 0.0
is_weekend 0.0
is_after_hours 0.0
date_only 0.0
log_source 0.0
dtype: float64
```

```
http
id      0.0
user    0.0
pc      0.0
url     0.0
content 0.0
timestamp 0.0
hour     0.0
day_of_week 0.0
is_weekend 0.0
is_after_hours 0.0
date_only 0.0
log_source 0.0
dtype: float64
```

User Activity Summary

```
[7]: summary = []
for name, df in logs.items():
    if "user" in df.columns:
        counts = df.groupby("user").size().rename(name)
        summary.append(counts)

summary_df = pd.concat(summary, axis=1).fillna(0)
summary_df["total"] = summary_df.sum(axis=1)
summary_df.sort_values("total", ascending=False).head()
```

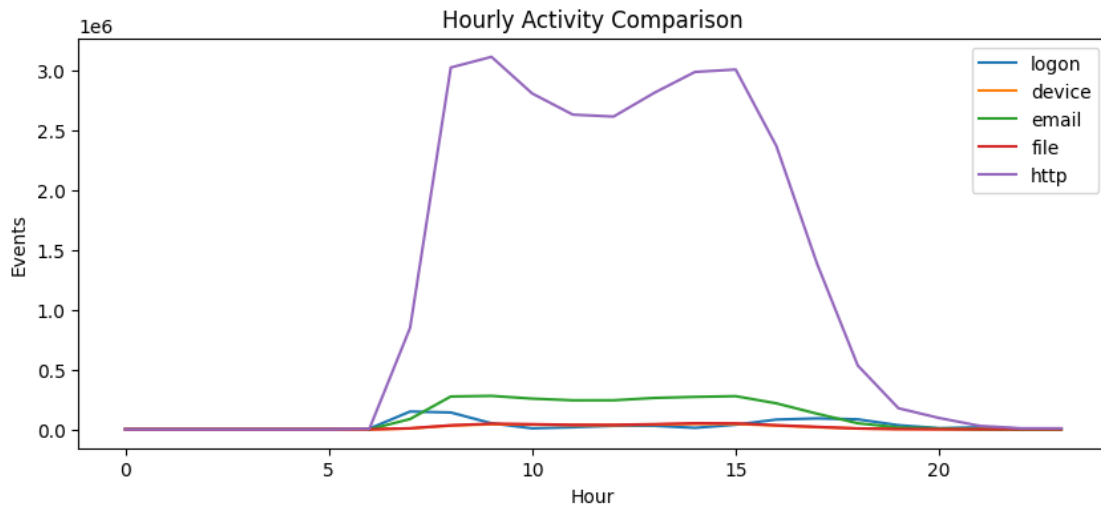
```
[7]:      logon  device  email    file    http    total
user
mss0001    960     0.0  12034     0.0  138000  150994.0
```

kbp0008	1197	0.0	9145	0.0	103800	114142.0
hcs0003	692	2006.0	9097	3045.0	98610	113450.0
d1m0051	1367	7533.0	7573	10673.0	82840	109986.0
hth0007	961	0.0	9116	0.0	98610	108687.0

Plot 1 — Hourly Distribution

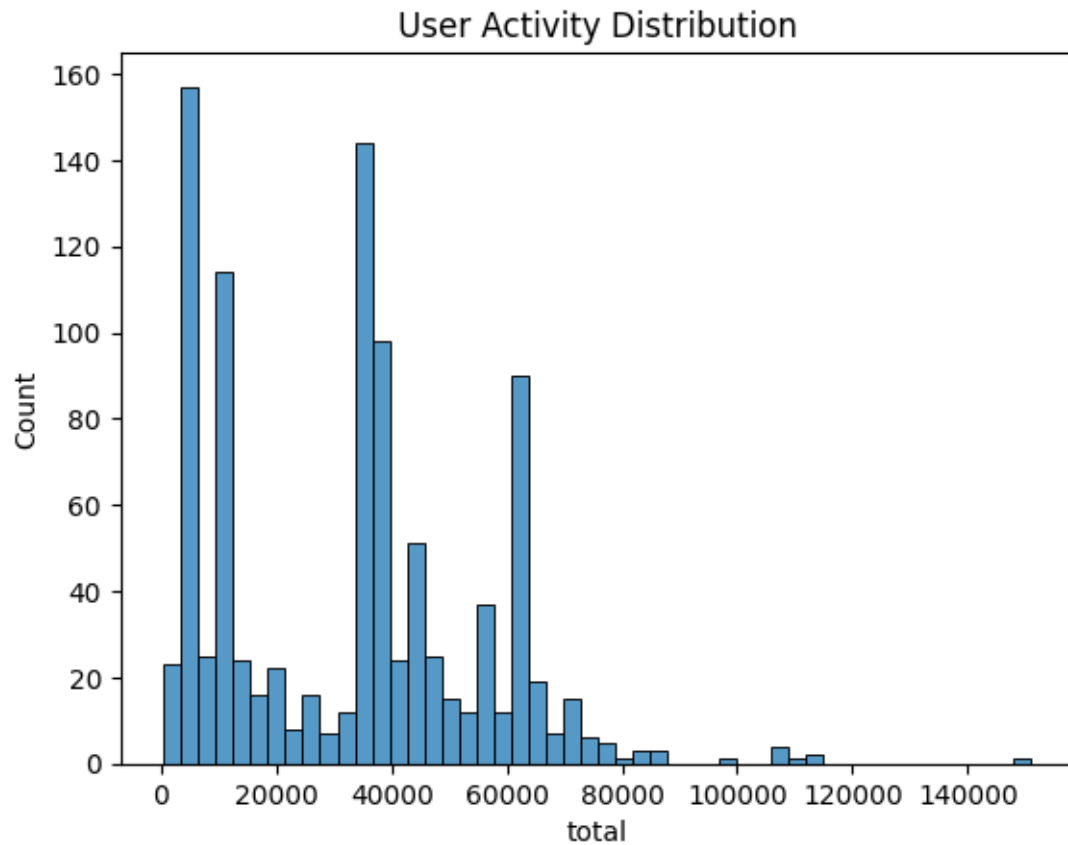
```
[8]: plt.figure(figsize=(10,4))
for name, df in logs.items():
    if "hour" in df.columns:
        df["hour"].value_counts().sort_index().plot(label=name)

plt.legend()
plt.title("Hourly Activity Comparison")
plt.xlabel("Hour")
plt.ylabel("Events")
plt.show()
```



Plot 2 — User Activity Distribution

```
[9]: sns.histplot(summary_df["total"], bins=50)
plt.title("User Activity Distribution")
plt.show()
```



After-hours Analysis

```
[10]: for name, df in logs.items():  
       if "is_after_hours" in df.columns:  
           print(name, df["is_after_hours"].mean() * 100)
```

```
logon 40.21739257585169  
device 8.398539641817553  
email 6.360507061082997  
file 8.394208909266778  
http 6.015282954748194
```